

EE/CE 354/361: Introduction to Probability and Statistics

Homework Assignment 3: Continuous Random Variables

Instructions

- **Attempt all questions.** Only **7 randomly chosen** questions will be graded.
- Show all working clearly.
- State assumptions where necessary.
- Total Points: **70**

Q1 (10 points)

You toss a biased coin two times. The random variable, X , represents the number of Heads obtained. Therefore, X is a Binomial Random Variable with parameter $n = 2$ with a probability of success, $p = 0.2$.

- a) Calculate and plot the PMF, $P_X(x)$ for the DRV X .

We now define a new discrete random variable Y , where the conditional PMF $P_{X|Y}(Y|X = x)$ is a discrete uniform distribution taking on values of x to 4.

(For example, if $X = 2$, then $Y|X = 2$ has a discrete Uniform distribution on $\{2, 3, 4\}$.)

- b) Calculate and plot the conditional PMF $P_{Y|X}(y|x)$ for all values of x .
- c) Calculate and plot the marginal PMF, $P_Y(y)$.
- d) Calculate and plot the Joint PMF, $P_{X,Y}(x, y)$.

Q2 (10 points)

If the height of Habib University students is modeled as a Normal random variable with mean of 65 inches and standard deviation of 3 inches, what is the probability that a randomly chosen HU student has a height of 6ft or more?

Q3 (10 points)

The time until the next flight departs from an airport follows the following distribution:

$$f(x) = \frac{1}{20} \text{ for } 25 \leq x \leq 45$$

- a) What is $E[X]$?
- b) What is the standard deviation of X ?
- c) Find the probability that the time until the departure of the next flight is at most 30 minutes.
- d) Find the probability that the time until the departure of the next flight is between 30 and 40 minutes.

Q4 (10 points)

Suppose that the number of kilometers you drive before you get into a car accident is modeled as an exponential random variable with a mean of 15,000 kms. You are planning to drive in your car from Karachi to Khunjerab Pass and back. What is the probability that you will make it back to Karachi without getting into an accident?

(Hint: According to Google Maps, the distance from Karachi to Khunjerab Pass is 2,100 km)

Q5 (10 points)

- a) Consider a javelin throw competition involving three athletes: Arshad, Babar, and Shaheen. Each athlete gets only one throw during the competition. Throws by each athlete are independent of the other athletes. Arshad can throw between 80m to 90m with all values being equally likely. Babar can throw between 85m to 90m with all values being equally likely. Shaheen can throw between 75m to 85m with all values being equally likely. Let random variable X be the distance (in meters) of the longest throw during the competition. What is the PDF of X ?
- b) X and Y are independent continuous random variables that are uniformly distributed with $X \sim U[0,2]$ and $Y \sim U[0,1]$. Let A be the event $\{0.25 \leq X \leq 0.5\}$. Find the following:
- i) $f_{X,Y}(1.5, 1.5)$
 - ii) $F_{X,Y}(x, y)$
 - iii) $E[X|A]$

Q6 (10 points)

- a) X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x, y) = \frac{xy^2}{81}$$

for $0 \leq x \leq 3$ and $0 \leq y \leq 3$. The joint PDF is zero for all other values of x and y . Are X and Y independent? Justify your answer.

- b) X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x, y) = x + cy^3$$

for $0 \leq x \leq 1$ and $0 \leq y \leq 1$. The joint PDF is zero for all other values of x and y .

- i) What is the value of constant c ?
- ii) Are X and Y independent? Justify your answer.

Q7 (10 points)

X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x, y) = \begin{cases} \frac{2xy^2}{81} & 0 \leq x \leq 3, \quad 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Let A be the event $\{0 \leq X \leq 1\}$. Find the following:

- i) $P(X \leq 1, Y \leq 1)$
- ii) $E[X|A]$

Q8 (10 points)

X and Y are independent continuous random variables that are uniformly distributed with $X \sim U[0, 2]$ and $Y \sim U[0, 2]$. Find the following:

- a) $f_{X,Y}(x, y)$
- b) $F_{X,Y}(1.5, 1.5)$
- c) $f_{Y|X}(y, 0.5)$
- d) $f_{X|Y}(0.5, 1)$
- e) $E[X|Y > 1.5]$

Q9 (10 points)

- a) Binary signal S is transmitted across a communication channel. The received signal is $Y = N + S$, where N is normal noise with zero mean and unit variance, independent of S . Let A be the event that the noise is between -0.25 and 0.25 (i.e. $N \in [-0.25, 0.25]$). What is $E[N|A]$? Justify your answer.
- b) Suppose you were told that the average odometer reading (number of kilometers a car has traveled in its lifetime so far) of all the red cars in Karachi was 55,000 km. Express this information in terms of two random variables X and Y . Define the random variables X and Y appropriately. (You can use any of the concepts applicable to two random variables such as joint PDF, conditional PDF, etc.)

Q10 (10 points)

Consider a promising new batter who performed really well in the PSL this year. He has been called up to the training camp of Team Pakistan and is likely to start his career as an international player in the near future. Based on his batting technique, a cricket expert has provided you the following information about his potential career as a batter for Team Pakistan: By the end of his international career, his batting average (number of runs scored per innings) inside Asia will be between 30.5 to 40.5 runs with all values being equally likely. His batting average outside Asia will be between 15.5 and 35.5 with all values being equally likely. Team Pakistan plays 40% of its matches outside Asia. Let X be the continuous random variable that captures his batting average at the end of his international career with Team Pakistan, what is the PDF of X ?

Q11 (10 points)

A city has two internet service providers, Charter and Xfinity, with the market share of 30% and 70% respectively. At any given time, the internet speed experienced by a customer of Charter is between 7Mbps and 15Mbps with all speeds being equally likely. At any given time, the internet speed experienced by a customer of Xfinity is between 10Mbps and 20Mbps with all speeds being equally likely. If an internet customer in the city is currently experiencing an internet speed of 12Mbps, what is the probability that the customer is served by Charter?

Q12 (10 points)

A binary signal S is transmitted and we are given that $P(S = 0) = 0.4$ and $P(S = 1) = 0.6$. The received signal is $Y = N + S$, where N is normal noise with zero mean and unit variance, independent of S . What is the probability that $S = 1$ if the observed value of received signal Y is 0.7.

Q13 (10 points)

On a sunny day, it takes a student between 30 to 45 minutes to get to Habib University campus with all times being equally likely. On a rainy day, it takes the same student between 40 to 60 minutes to get to campus with all times being equally likely. Assume a day is sunny with probability 0.75 and rainy with probability 0.25.

- a) Find the PDF of the time T that it takes the student to get to campus.
- b) On a given day, it took the student 42 minutes to get to campus. What is the probability that this particular day was rainy?

Q14 (10 points)

Consider a Bernoulli random variable X with parameter $p = 0.1$. We observe X in the presence of additive noise $N \sim U[-2, 2]$. As a result, our observation is random variable $Y = X + N$. If our observation is 0.8, what is the "most likely" value of X ?

(**Note:** Provide a justification of your answer based on the various theorems/rules/laws/results discussed in this course. Only intuitive argument is not acceptable for this question and will be given no partial credit)

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.5279	0.53188	0.53586
0.1	0.53983	0.5438	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.6293	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.6591	0.66276	0.6664	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.7054	0.70884	0.71226	0.71566	0.71904	0.7224
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.7549
0.7	0.75804	0.76115	0.76424	0.7673	0.77035	0.77337	0.77637	0.77935	0.7823	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.8665	0.86864	0.87076	0.87286	0.87493	0.87698	0.879	0.881	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.9032	0.9049	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.9222	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.9452	0.9463	0.94738	0.94845	0.9495	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.9608	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.9732	0.97381	0.97441	0.975	0.97558	0.97615	0.9767
2	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.9803	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.983	0.98341	0.98382	0.98422	0.98461	0.985	0.98537	0.98574
2.2	0.9861	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.9884	0.9887	0.98899
2.3	0.98928	0.98956	0.98983	0.9901	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.9918	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.9943	0.99446	0.99461	0.99477	0.99492	0.99506	0.9952
2.6	0.99534	0.99547	0.9956	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.9972	0.99728	0.99736
2.8	0.99744	0.99752	0.9976	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.999
3.1	0.99903	0.99906	0.9991	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.9994	0.99942	0.99944	0.99946	0.99948	0.9995
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.9996	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.9997	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.9998	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.9999	0.9999	0.9999	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4	0.99997	0.99997	0.99997	0.99997	0.99997	0.99997	0.99998	0.99998	0.99998	0.99998

Table 1: The standard normal table